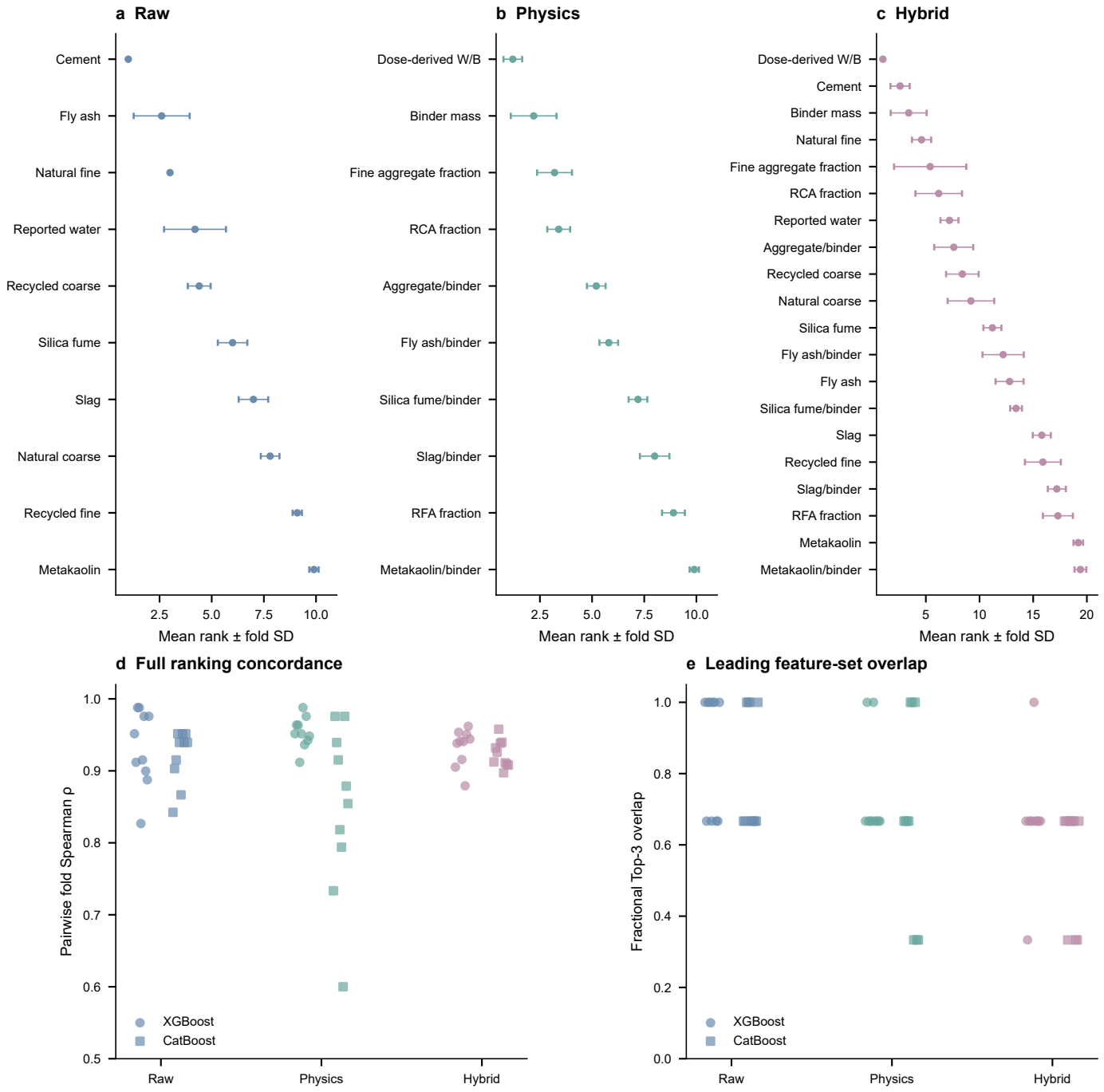

a Raw
Cement
Fly ash
Natural fine
Reported water
Recycled coarse
Silica fume
Slag
Natural coarse
Recycled fine
Metakaolin
2.5
5.0
7.5
10.0
Mean rank ± fold SD
b Physics
Dose-derived W/B
Binder mass
Fine aggregate fraction
RCA fraction
Aggregate/binder
Fly ash/binder
Silica fume/binder
Slag/binder
RFA fraction
Metakaolin/binder
2.5
5.0
7.5
10.0
Mean rank ± fold SD
c Hybrid
Dose-derived W/B
Cement
Binder mass
Natural fine
Fine aggregate fraction
RCA fraction
Reported water
Aggregate/binder
Recycled coarse
Natural coarse
Silica fume
Fly ash/binder
Fly ash
Silica fume/binder
Slag
Recycled fine
Slag/binder
RFA fraction
Metakaolin
Metakaolin/binder
5
10
15
20
Mean rank ± fold SD
d Full ranking concordance
Pairwise fold Spearman ρ
1.0
0.9
0.8
0.7
0.6
0.5
XGBoost
CatBoost
Raw
Physics
Hybrid
e Leading feature-set overlap
Fractional Top-3 overlap
1.0
0.8
0.6
0.4
0.2
0.0
XGBoost
CatBoost
Raw
Physics
Hybrid